

Ahmed Mohamed Abdel Fattah

Chemical & petrochemical Engineer

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📍 Ahmed Mohamed Abdel Fattah

Dammam, Saudi Arabia

Summery

Motivated Chemical & Petrochemical Engineering graduate with hands-on experience in process design, simulation, and plant operations. Eager to learn from experts, enhance technical skills, and contribute to innovative engineering solutions in a dynamic team environment.

Education

Bachelor Degree in Petrochemical Engineering, Pharos University, Egypt.

2019 - 2024

GPA: 3.77 / 4 Ranked 3rd

Graduation Project: Design of Ethane-Propane Recovery Process from Natural Gas at Abu QIR petroleum company.

Project Grade: Excellent

Projects

Graduation project

Abu Qir plant

Oct 2023-May2024

- Defined the project's design basis and technical specifications.
- Generated and screened multiple conceptual design options using feasibility criteria.
- Selected appropriate process units based on feed composition and product specifications.
- Developed detailed process flow diagrams (PFDs) and performed heat and material balances.
- Simulated the process using Aspen HYSYS to optimize efficiency and performance.
- Created piping and instrumentation diagrams (P&IDs) with integrated control and safety systems.
- Conducted HAZOP studies and cause-and-effect analysis to assess and mitigate process risks.
- Evaluated economic feasibility to support technical decisions and design justification.

Industrial Internships

Rashid Petroleum company (Shell-JV)

Idku, Egypt

Aug 2022 & Sep 2023

- Gained theoretical understanding of gas dehydration & sweetening processes.
- Developed an understanding of cause-and-effect relationships in process systems and gained exposure to permit-to-work (PTW) processes to ensure safety and operational efficiency.

General Petroleum Company (GPC)

Red Sea, Egypt

Aug 2023

- Observed DCS systems and basic process control operations.
- Understood primary crude oil treatment steps and offshore platform components.

Alexandria National Refining & Petrochemicals Co. (ANRPC)

Alexandria, Egypt

Jul 2023

- Studied catalytic reforming processes including CCR, Towers, and Furnace.
- Gained awareness of reactor operations and regeneration principles.

Abu Qir Fertilizers and Chemical Industries Company

Alexandria, Egypt

Aug 2022

- Learned about ammonia and urea production through theoretical exposure to steam reforming and water-gas shift processes.

Sidi Kerir Petrochemicals Company (SIDPEC)

Alexandria, Egypt

Aug 2022

- Explored process concepts related to steam generation, feed preparation, steam cracking.
- Gained a theoretical overview quenching processes, compression, drying techniques, chilling

trains, and polymerization.

Holding Company for Water and Wastewater Treatment Alexandria, Egypt Sep 2021

- Learned about water treatment stages such as coagulation, flocculation, sedimentation, filtration, and disinfection.

The Egyptian Ethylene & Derivatives Company (ETHYDCO) Alexandria, Egypt Aug 2021

- Reviewed major plant systems including feed handling, process flow diagrams, and heat exchanger roles.
- Gained theoretical insights into P&IDs and process utility operations.

Courses

Process & Operations

- Introduction to Process Design Engineering Activities – Booststrand
- Process Plant Operator Induction – Udemy
- Equipment: Operation and Troubleshooting – Online Course
- Fundamentals of Water Treatment – Horizon Academy

Process Simulation & Optimization

- Process Simulation using Aspen HYSYS – Online Course
- Lean Six Sigma White Belt – Six Sigma Council

Safety & Compliance

- Fire Safety Management – ITCILO
- OHS Training Program (OSHA General Standards) – Prosafe
- OHS Training Program (OSHA Construction Standards) – Prosafe

Skills

Technical Skills

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|---------------------------|--------------------------------|--------------------------|
| - Process Control | - Unit Operations & process | - Heat & Mass Balances |
| - PFD & P&ID Reading | - Aspen HYSYS (Certified User) | - AutoCAD (Basic) |
| - Cause & Effect Analysis | - Automatic control | - Basic design knowledge |

Soft Skills

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| - Communication skills | - Active listening | - Problem-solving abilities |
| - Team Collaboration | - Adaptability | - Attention to Detail |

Awards

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| - Academic Excellence Award – Faculty of Engineering, Pharos University | June 2024 |
| - Exemplary Student Recognition – Faculty of Engineering, Pharos University | May 2023 |
| - Academic Excellence Award – Faculty of Engineering, Pharos University | January 2023 |

Language

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| - Arabic: Mother Tongue | - English: Very good |
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Additional information

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| - Marital Status: Single | - Birth Date: June 14, 2000 |
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